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Ising Model

Monte Carlo Simulation of the Two-Dimensional Ising Model

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1 The Two-Dimensional Ising Model

1.1 Spin Lattice

The system consists of a square lattice of size

$$N \times N.$$

Each lattice point contains one spin variable $s_{i,j}$, where

$$s_{i,j} \in \{-1, +1\}.$$

The total number of spins is therefore

$$N_{\text{spins}} = N^2.$$

A spin interacts only with its nearest neighbours. In a two-dimensional square lattice, each spin has four nearest neighbours:

$$s_{i-1,j}, \quad s_{i+1,j}, \quad s_{i,j-1}, \quad s_{i,j+1}.$$

In this simulation, periodic boundary conditions are used. This means that the lattice wraps around itself. Therefore, the spin at the right edge has as its right neighbour the spin at the left edge, and the spin at the top edge has as its upper neighbour the spin at the bottom edge. Periodic boundary conditions reduce finite-size boundary effects and make the finite lattice behave more like an infinite system.

1.2 Hamiltonian

The energy of a spin configuration is described by the Ising Hamiltonian:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i.$$

The notation $\langle i, j \rangle$ indicates that the sum is taken over unique nearest-neighbour pairs. The parameter J is the interaction strength, and h is the external magnetic field.

For this assignment, the material is assumed to be ferromagnetic, so

$$J = 1.$$

The Hamiltonian contains two parts:

$$H = E_{\text{int}} + E_{\text{field}},$$

where

$$E_{\text{int}} = -J \sum_{\langle i,j \rangle} s_i s_j$$

is the interaction energy, and

$$E_{\text{field}} = -h \sum_i s_i$$

is the magnetic-field energy.

1.3 Physical Meaning of the Interaction Term

For a pair of neighbouring spins, the interaction energy is

$$E_{ij} = -Js_i s_j.$$

Since $J > 0$, aligned spins lower the energy. For example, if

$$s_i = +1, \quad s_j = +1,$$

then

$$E_{ij} = -J(+1)(+1) = -J.$$

Similarly, if

$$s_i = -1, \quad s_j = -1,$$

then

$$E_{ij} = -J(-1)(-1) = -J.$$

In both cases, the contribution to the energy is negative.

On the other hand, if the two spins are opposite,

$$s_i = +1, \quad s_j = -1,$$

then

$$E_{ij} = -J(+1)(-1) = +J.$$

Thus, opposite spins increase the energy. Therefore, for $J > 0$, the system energetically favors neighboring spins that are aligned. This is the microscopic origin of ferromagnetic ordering.

1.4 Physical Meaning of the External Field Term

The external field contribution is

$$E_{\text{field}} = -h \sum_i s_i.$$

If $h > 0$, then spins with value $+1$ lower the energy:

$$-h(+1) = -h.$$

Spins with value -1 increase the energy:

$$-h(-1) = +h.$$

Therefore, a positive external magnetic field favors positive spins. If the field is strong enough, it can force the system to remain magnetized even at relatively high temperatures.

In this assignment, we will study three configurations:

$$h = 0,$$

$$h > 0,$$

and

$$h > 1.$$

These regimes correspond to no external field, a weak external field, and a strong external field.

2 Statistical Mechanics Background

2.1 Boltzmann Distribution

At thermal equilibrium, a configuration with energy E occurs with probability proportional to the Boltzmann factor:

$$P(E) \propto e^{-E/(k_B T)}.$$

In this assignment, Boltzmann's constant is taken to be

$$k_B = 1.$$

Therefore, the Boltzmann factor becomes

$$P(E) \propto e^{-E/T}.$$

This expression shows that low-energy configurations are more probable than high-energy configurations. However, high-energy configurations are not impossible, especially at high temperature.

At low temperature, the factor $e^{-E/T}$ strongly favors low-energy configurations. Therefore, the system tends to become ordered. At high temperature, thermal fluctuations become stronger, and disordered configurations become more likely.

2.2 Need for Monte Carlo Methods

For a lattice of size $N \times N$, the total number of possible spin configurations is

$$2^{N^2}.$$

For even a moderate lattice size, this number becomes extremely large. For example, if

$$N = 30,$$

then the number of possible configurations is

$$2^{900}.$$

It is impossible to calculate the thermodynamic properties by explicitly summing over all possible configurations. Monte Carlo methods solve this problem by sampling a representative subset of configurations according to their statistical weight.

Instead of visiting every possible state, the Metropolis algorithm generates a Markov chain of configurations that, after sufficient time, samples the Boltzmann distribution [1, 2].

3 Metropolis Algorithm

3.1 Basic Idea

The Metropolis algorithm is a single-spin-flip Monte Carlo method. Starting from an initial spin configuration, the algorithm repeatedly selects a spin and proposes to flip it:

$$s_i \rightarrow -s_i.$$

The energy change caused by this flip is

$$\Delta E = E_{\text{new}} - E_{\text{old}}.$$

The proposed spin flip is accepted according to the following rule:

If $\Delta E < 0$, accept the flip.

If the new configuration has higher energy,

$$\Delta E > 0,$$

then the flip is accepted with probability

$$P_{\text{accept}} = e^{-\Delta E/T}.$$

This rule allows the system to move toward low-energy states while still allowing occasional transitions to higher-energy states. These higher-energy moves are necessary because physical systems at nonzero temperature experience thermal fluctuations.

3.2 Derivation of the Local Energy Change

When only one spin s_i is flipped, only the energy terms involving that spin change. Therefore, it is not necessary to recalculate the full Hamiltonian after every proposed move.

The local energy of spin s_i before flipping is

$$E_{\text{old}} = -Js_i \sum s_j - hs_i,$$

where the sum is over the nearest neighbors of s_i .

After flipping,

$$s_i \rightarrow -s_i,$$

the new local energy is

$$E_{\text{new}} = -J(-s_i) \sum s_j - h(-s_i).$$

This becomes

$$E_{\text{new}} = Js_i \sum s_j + hs_i.$$

Therefore,

$$\Delta E = E_{\text{new}} - E_{\text{old}}.$$

Substituting the two expressions gives

$$\Delta E = \left(Js_i \sum s_j + hs_i \right) - \left(-Js_i \sum s_j - hs_i \right).$$

Thus,

$$\Delta E = 2Js_i \sum s_j + 2hs_i.$$

Factoring out $2s_i$, we obtain

$$\Delta E = 2s_i (J_j + h).$$

This is the formula used in the Metropolis update.

3.3 Role of Temperature

The acceptance probability for an energy-increasing move is

$$P_{\text{accept}} = e^{-\Delta E/T}.$$

At low temperature, T is small, so for positive ΔE , the ratio $\Delta E/T$ is large. Therefore,

$$e^{-\Delta E/T}$$

is very small. As a result, energy-increasing moves are rarely accepted.

At high temperature, T is large, so $\Delta E/T$ is smaller. Therefore, energy-increasing moves are more likely to be accepted. This produces stronger fluctuations and a more disordered spin configuration.

3.4 Monte Carlo Sweep

A single proposed spin flip updates only one spin. A Monte Carlo sweep consists of

$$N^2$$

attempted spin flips. Since the lattice contains N^2 spins, one sweep corresponds to approximately one attempted update per spin.

In the implementation, the selected spin positions are chosen randomly. Therefore, during one sweep, some spins may be selected more than once, while others may not be selected. However, on average, one sweep corresponds to N^2 update attempts.

4 Simulation Procedure

4.1 Initialization

The simulation begins by constructing a square lattice of size

$$N \times N.$$

The initial spin configuration can be chosen in different ways:

$$s_{i,j} = +1 \quad \text{for all sites,}$$

$$s_{i,j} = -1 \quad \text{for all sites,}$$

or randomly,

$$s_{i,j} = \pm 1.$$

A random initial condition is often useful because it does not impose an initially ordered state on the system. However, at low temperature, the system will usually evolve toward an ordered state after sufficient thermalization.

4.2 Thermalization

The initial lattice configuration is generally not an equilibrium configuration. Therefore, the system must first be allowed to evolve before measurements are taken. This stage is called thermalization or equilibration.

During thermalization, the Metropolis sweep is applied repeatedly, but no measurements are recorded. The purpose is to allow the system to lose memory of the initial condition and approach thermal equilibrium.

If N_{therm} is the number of thermalization sweeps, then the thermalization stage is

repeat Metropolis sweep N_{therm} times.

The number of required thermalization sweeps depends on the lattice size, temperature, and magnetic field. Near the critical temperature, equilibration may become slower because of critical slowing down [2].

4.3 Measurement Stage

After thermalization, measurements are recorded for a number of additional sweeps. If the number of measurement sweeps is N_{meas} , then after each sweep the energy and magnetization are calculated.

The total magnetization is

$$M = \sum_i s_i.$$

The average spin per site is

$$m = \frac{M}{N^2}.$$

The energy per spin is

$$e = \frac{E}{N^2}.$$

The measured quantities are then averaged over all measurement sweeps:

$$\langle e \rangle = \frac{1}{N_{\text{meas}}} \sum_{t=1}^{N_{\text{meas}}} e_t,$$

and

$$\langle m \rangle = \frac{1}{N_{\text{meas}}} \sum_{t=1}^{N_{\text{meas}}} m_t.$$

For zero external field, the system has symmetry between positive and negative magnetization. Therefore, at low temperature, the system may settle into either a mostly positive or mostly negative state. In this case, it is often useful to measure the absolute magnetization,

$$|m| = \frac{|M|}{N^2},$$

because it measures the degree of magnetic order independently of direction.

5 Energy Calculation

5.1 Avoiding Double Counting

The Hamiltonian contains the nearest-neighbour sum

$$\sum_{\langle i,j \rangle} s_i s_j.$$

The notation $\langle i, j \rangle$ means that each nearest-neighbour pair is counted once. In the numerical implementation, each spin has four neighbours, but if all four neighbours are counted for every spin, every pair is counted twice.

For example, if spin A is a neighbour of spin B , then counting all neighbours for every spin includes both interactions

$$A \rightarrow B$$

and

$$B \rightarrow A.$$

These are the same physical bond. Therefore, the interaction energy would be double counted.

There are two equivalent ways to calculate the total interaction energy.

The first method is to count only the right and down neighbours:

$$E_{\text{int}} = -J \sum_{i,j} s_{i,j} (s_{i,j+1} + s_{i+1,j}).$$

This counts every horizontal and vertical bond exactly once.

The second method is to count all four neighbours and divide the interaction energy by two:

$$E_{\text{int}} = -\frac{J}{2} \sum_{i,j} s_{i,j} (s_{i-1,j} + s_{i+1,j} + s_{i,j-1} + s_{i,j+1}).$$

Both approaches give the same total energy. In this work, the right-and-down method is preferred because it avoids the need for the factor $1/2$.

5.2 Total Energy

Using the right-and-down method, the total energy is calculated as

$$E = -J \sum_{i,j} s_{i,j} (s_{i,j+1} + s_{i+1,j}) - h \sum_{i,j} s_{i,j}.$$

With periodic boundary conditions, the indices are understood modulo N . Therefore,

$$s_{i,N} = s_{i,0}$$

and

$$s_{N,j} = s_{0,j}.$$

This ensures that the right neighbour of the last column is the first column, and the down neighbour of the last row is the first row.

6 Observables

6.1 Magnetization

The total magnetization is defined as

$$M = \sum_i s_i.$$

The normalized magnetization, or average spin per site, is

$$m = \frac{M}{N^2}.$$

The value of m lies between -1 and $+1$:

$$-1 \leq m \leq 1.$$

If

$$m \approx 1,$$

then most spins are $+1$. If

$$m \approx -1,$$

then most spins are -1 . If

$$m \approx 0,$$

then the system is disordered and contains approximately equal numbers of positive and negative spins.

6.2 Energy Per Spin

The normalized energy is

$$e = \frac{E}{N^2}.$$

This quantity is useful because it allows systems of different lattice sizes to be compared. A more negative energy usually indicates a more ordered configuration, because neighbouring spins are more often aligned.

6.3 Average Quantities

During the measurement stage, the simulation records values

$$e_1, e_2, \dots, e_{N_{\text{meas}}},$$

and

$$m_1, m_2, \dots, m_{N_{\text{meas}}}.$$

The average energy per spin is

$$\langle e \rangle = \frac{1}{N_{\text{meas}}} \sum_{t=1}^{N_{\text{meas}}} e_t.$$

The average magnetization per spin is

$$\langle m \rangle = \frac{1}{N_{\text{meas}}} \sum_{t=1}^{N_{\text{meas}}} m_t.$$

For zero magnetic field, the average absolute magnetization may also be used:

$$\langle |m| \rangle = \frac{1}{N_{\text{meas}}} \sum_{t=1}^{N_{\text{meas}}} |m_t|.$$

7 Algorithmic Description

The Metropolis simulation can be summarized as follows:

1. Choose the lattice size N , temperature T , coupling constant J , and external field h .
2. Initialize the lattice spins randomly or in an ordered state.
3. Repeat the Metropolis sweep for a fixed number of thermalization sweeps.
4. After thermalization, perform measurement sweeps.
5. After each measurement sweep, calculate the energy E and magnetization M .
6. Normalize the measured quantities:

$$e = \frac{E}{N^2}, \quad m = \frac{M}{N^2}.$$

7. Average the measured values over all measurement sweeps.
8. Repeat the simulation for different temperatures and magnetic fields.

A single spin-flip attempt is described by the following steps:

1. Select a random lattice site (i, j) .
2. Calculate the sum of the four nearest neighbors:

$$S_{\text{nn}} = s_{i-1,j} + s_{i+1,j} + s_{i,j-1} + s_{i,j+1}.$$

3. Calculate the energy change:

$$\Delta E = 2s_{i,j} (JS_{\text{nn}} + h).$$

4. If $\Delta E < 0$, accept the flip.
5. If $\Delta E > 0$, accept the flip with probability:

$$P_{\text{accept}} = e^{-\Delta E/T}.$$

6. If the move is accepted, update:

$$s_{i,j} \rightarrow -s_{i,j}.$$

8 Simulation Parameters

In this assignment, the interaction strength is fixed as

$$J = 1.$$

The Boltzmann constant is also set to

$$k_B = 1.$$

A possible set of temperatures is

$$T = 0.5, \quad T = 2.5, \quad T = 4.0.$$

These values are useful because they sample three different regimes:

$$T < T_c,$$

$$T \approx T_c,$$

and

$$T > T_c.$$

Since

$$T_c \approx 2.269,$$

the value $T = 0.5$ corresponds to the ordered low-temperature phase, $T = 2.5$ is close to the critical region, and $T = 4.0$ corresponds to the disordered high-temperature phase.

The external magnetic fields may be chosen as

$$h = 0,$$

$$h = 0.5,$$

and

$$h = 2.0.$$

These represent no external field, a weak positive field, and a strong positive field. An example parameter set is shown in Table 1.

Table 1: Example simulation parameters.

Parameter	Value
Lattice size	$N = 40$
Coupling constant	$J = 1$
Boltzmann constant	$k_B = 1$
Temperatures	0.5, 2.5, 4.0
External fields	0, 0.5, 2.0
Thermalization sweeps	100
Measurement sweeps	10000
Boundary conditions	Periodic

9 Data Analysis and Plots

The main quantities plotted in this assignment are:

1. Average energy per spin versus temperature:

$$\langle e \rangle \text{ vs } T.$$

2. Average magnetization per spin versus temperature:

$$\langle m \rangle \text{ vs } T.$$

For each plot the spins on lattice have been initialized with three configuration. First configuration is all spins are up, second all down and third randomly. The temperature-dependent plots show how the thermodynamic behavior changes with temperature.

9.1 Average Energy Versus Temperature

The average energy per spin is expected to increase with temperature. At low temperature, neighboring spins are mostly aligned, so the interaction energy is strongly negative. At high temperature, the spin configuration becomes more disordered, and fewer neighboring pairs are aligned. Therefore, the energy becomes less negative.

A plot of $\langle e \rangle$ versus T should therefore show an increasing trend.

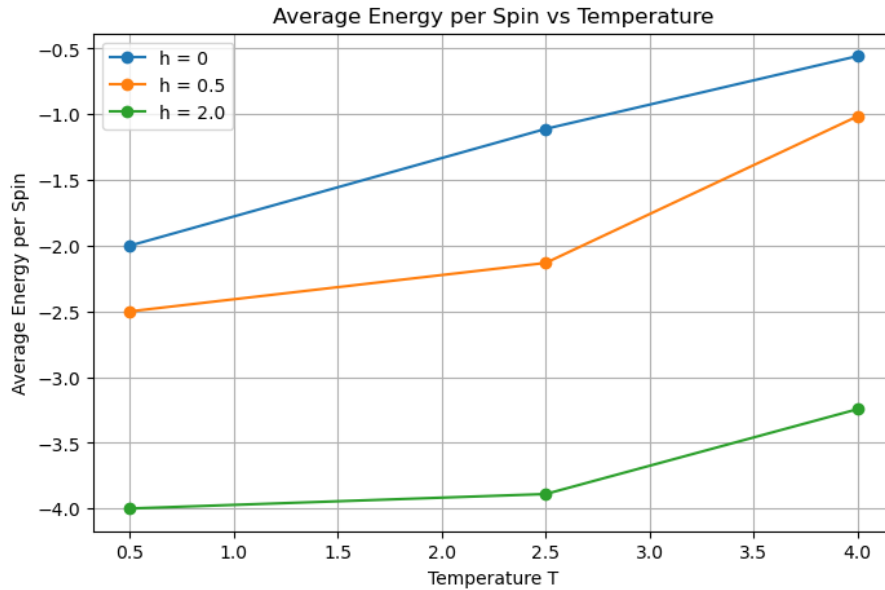


Figure 1: Average energy per spin as a function of temperature for different values of the magnetic field. The spins on initialize lattice are **up**

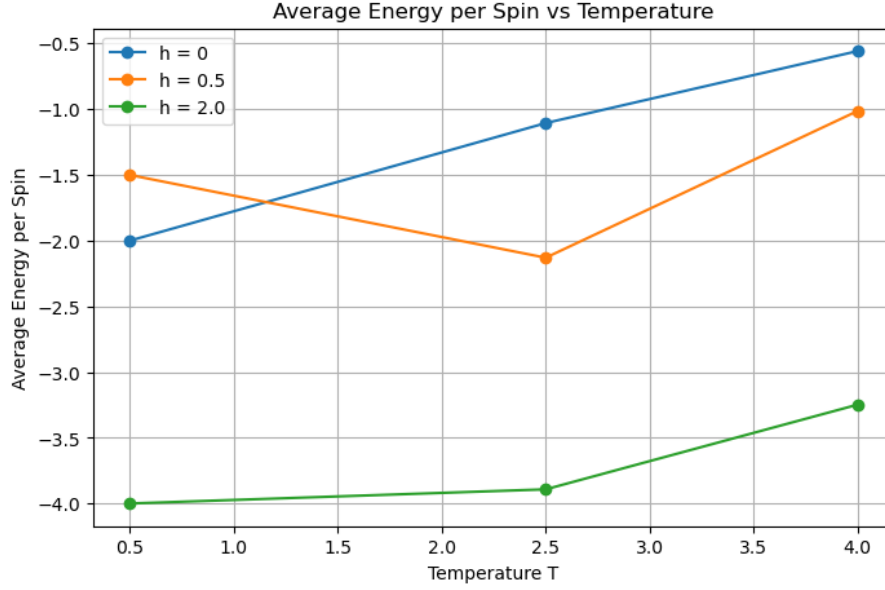


Figure 2: Average energy per spin as a function of temperature for different values of the magnetic field. The spins on initialize lattice are ***down***

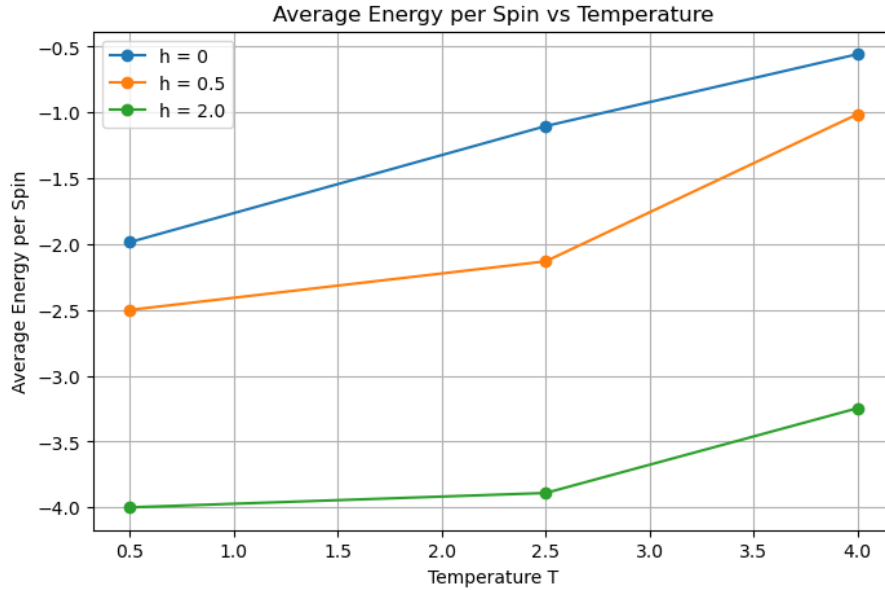


Figure 3: Average energy per spin as a function of temperature for different values of the magnetic field. The spins on initialize lattice are ***random***

9.2 Average Magnetization Versus Temperature

The average magnetization per spin measures the degree of magnetic ordering. For zero field, the magnetization is large in magnitude below the critical temperature and approaches zero above the critical temperature.

For positive external field, the magnetization remains positive because the field favors +1 spins. The stronger the field, the larger the magnetization remains at high temperature.

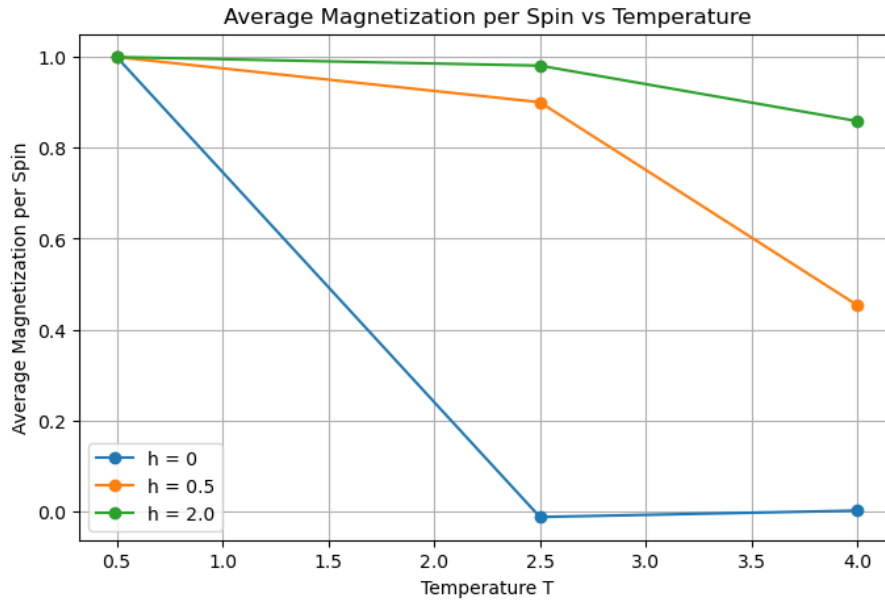


Figure 4: Average magnetization per spin as a function of temperature for different values of the magnetic field. The spins on initialize lattice are *up*

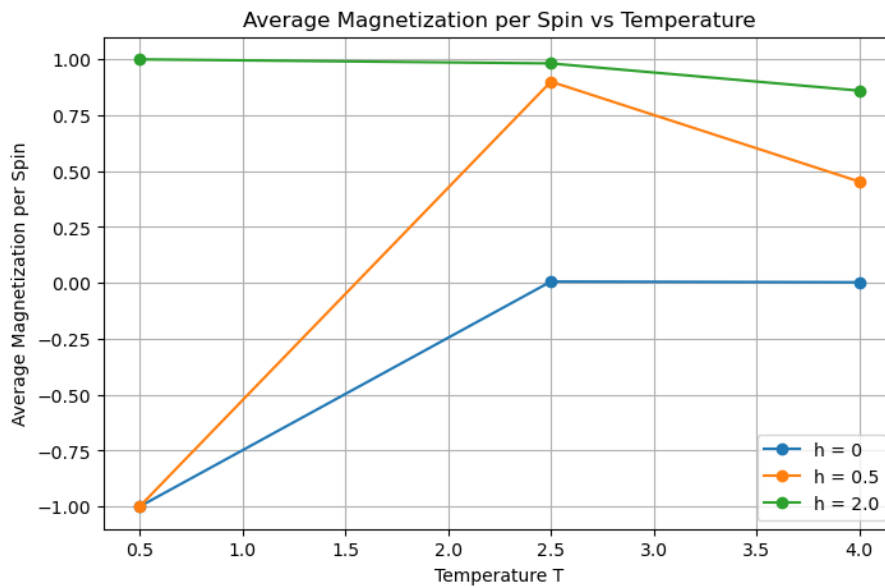


Figure 5: Average magnetization per spin as a function of temperature for different values of the magnetic field. The spins on initialize lattice are *down*

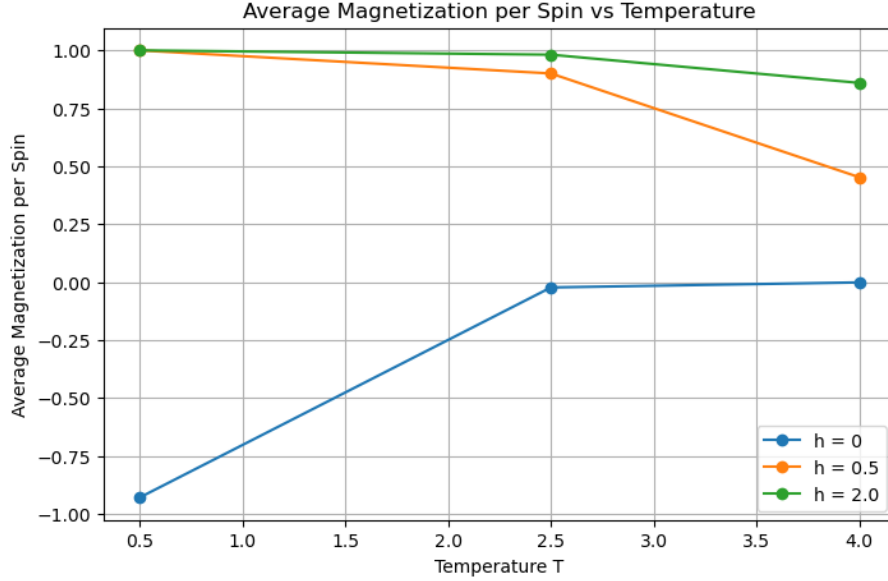


Figure 6: Average magnetization per spin as a function of temperature for different values of the magnetic field. The spins on initialize lattice are *random*

10 Discussion

The Metropolis Monte Carlo simulation reproduces the expected qualitative behavior of the two-dimensional ferromagnetic Ising model. At low temperatures, the system tends to minimize its energy by aligning neighboring spins. This produces an ordered phase with large magnetization. At high temperatures, thermal fluctuations become stronger and destroy the long-range magnetic order.

The effect of the external magnetic field is also clear. When $h = 0$, the system has no preference for positive or negative magnetization. Therefore, at low temperature, it may become ordered in either direction. When $h > 0$, positive spins are favored, and the magnetization becomes biased toward positive values. For strong external field, the field dominates over thermal disorder, so the system remains strongly magnetized even at high temperature.

The energy measurements also behave consistently with physical expectations. At low temperature, aligned neighboring spins give a strongly negative interaction energy. As temperature increases, more spin pairs become misaligned, and the average energy per spin becomes less negative.

11 Conclusion

In this work, the two-dimensional Ising model was studied using the Metropolis Monte Carlo algorithm. The lattice consisted of binary spins interacting with their nearest neighbors, with periodic boundary conditions used to reduce boundary effects. The Metropolis algorithm was used to sample spin configurations according to the Boltzmann distribution.

The simulation shows that temperature and external magnetic field strongly affect the magnetic behavior of the system. At low temperature, the system becomes ordered and develops strong magnetization. At high temperature, thermal fluctuations disorder the

system and reduce the magnetization. A positive external field favors positive spins, while a strong field can maintain magnetic order even at higher temperatures.

Overall, the Ising model provides a simple but powerful example of how microscopic interactions can produce macroscopic phase behavior. The Monte Carlo method is especially useful because it allows the statistical properties of the system to be estimated without enumerating all possible spin configurations.

12 Python Code

All code used to generate the results, figures, and simulations in this report is publicly available at:

[Project GitHub Repository](#)

References

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- [2] D. P. Landau and K. Binder, *A Guide to Monte Carlo Simulations in Statistical Physics*, Cambridge University Press, Cambridge, 2005.